

FORECAST ANALYTICS

Excel Exercise 1 — Explore Demand

Insert → PivotTable

Set:

Rows: Month

Values:

- Sum of Demand
- Sum of Forecast

Questions:

“What do you notice before calculating anything?”

Understand following topics: **Trend, Seasonality, Gap, Underforecast, Volatility**

Excel Exercise 2 — Actual vs Forecast Chart

Select: Month | Actual Demand | Forecast

Then: **Insert → Line Chart**

Create two lines:

- **Actual Demand**
- **Forecast Demand**

Question: “What does the gap between these lines represent?” **Forecast Error**

Forecast Error

Use: **Error: Actual – Forecast**

Positive error under this convention means: **UNDERFORECAST** Demand was higher than expected.

Business consequences could include: **Stockout ↑ OTIF ↓ Expedited freight ↑ Lost sales ↑**

Four Forecast Metrics

METRIC	QUESTION IT ANSWERS	TITLE
MAE	How many units are we wrong by?	Mean Absolute Error
MAPE	On average, what % are we wrong by?	Mean Average % Error
WAPE	How large is total error relative to total demand?	Weighted Average % Error
BIAS	Are we systematically forecasting high or low?	Bias

MAE

Suppose the new Absolute Error column is Y. =AVERAGE(Y2:Y1801)

Explanation: “Our forecast misses actual demand by about 18 units per transaction on average.”

Excel Exercise 3 — MAPE

If you created Absolute Percentage Error: =AVERAGE(Z2:Z1801)

Suppose result: **14.8%**

We may conclude: “Forecast accuracy = 85.2%.” Organizations sometimes express $1 - \text{MAPE}$ as an intuitive “forecast accuracy,” but they should always be clear about the definition being used.

Excel Exercise 4 — WAPE

WAPE = Total Absolute Error / Total Actual Demand

If the result is: 0.126, then: **WAPE = 12.6%**

An intuitive WAPE-based accuracy expression would be: **87.4%** ($1 - \text{WAPE}$). Again, label it clearly as **WAPE-based forecast accuracy**.

Excel Exercise 5 — Bias (Forecast – Actual)

Let's say: **Bias = -8%**

“What does this mean?” The forecasting process is systematically **underforecasting**.

Underforecast

↓

Insufficient inventory

↓

Stockouts

↓

Expediting

↓

OTIF deterioration

PART 2

Create another PivotTable.

Rows

- Category

Values

- Sum of Demand
- Sum of Forecast
- Sum of Absolute Error

Then add: Region or SKU

Which categories/SKUs are driving forecast error?

Add Conditional Formatting

Create a summary table:

SKU	Actual	Forecast	Error %
SKU-101	1,250	1,190	4.8%
SKU-102	850	610	28.2%
SKU-103	2,100	2,180	3.8%
SKU-104	430	620	44.2%

Select Error %: **Home → Conditional Formatting → Color Scales**

“Would you investigate all 150 SKUs?” No. Focus on exceptions.

Excel Exercise 6 — Moving Average Forecast

Introducing a simple forecasting method. Create a separate monthly summary:

Month	Actual Demand	3-Month Moving Average
Jan	10,000	
Feb	11,000	
Mar	12,000	
Apr	13,000	11,000
May	14,000	12,000
Jun	16,000	13,000

Moving average is average of previous k number of periods. **“What does moving average do?”** It smooths short-term fluctuations.

“What is its weakness?” It **lags** when demand is rapidly increasing or decreasing.

Excel Exercise 7 — Excel Forecast Sheet

Create a clean monthly series: Month & Demand. Then: **Data → Forecast Sheet**

Excel can generate a forecast using its exponential-smoothing functionality.

It shows: Historical Demand, Forecast, Upper Confidence Bound and Lower Confidence Bound

Learning Forecast ≠ certainty. Forecasting should communicate uncertainty.

Forecasting: From basic to advance: Moving Average → Exponential Smoothing → ML

LEVEL 1: Naïve / Historical Average

LEVEL 2: Moving Average

LEVEL 3: Exponential Smoothing

LEVEL 4: Time-Series Models

LEVEL 5: Machine Learning

LEVEL 6: Demand Sensing / AI

But Remember: More sophisticated ≠ automatically more accurate.

What Data Could Improve the Forecast?

- Historical Sales
- Promotions
- Price
- Festival/holiday
- Customer orders
- Sales pipeline
- Weather
- Economic indicators
- Product launches
- Competitor activity
- Stock availability
- Supplier constraints

Understanding the concept of **demand sensing**.

Final Excel Challenge: Answer based on your analysis

THE COO WANTS THREE ANSWERS

1. Which product category has the poorest forecast performance?
2. Are we systematically overforecasting or underforecasting?
3. Which five SKUs deserve planner attention?

You may recommend only 3 actions. Work individually and discuss with your friend.

Possible recommendations might involve:

forecast-model adjustment, planner review, safety-stock review, supplier mitigation, customer collaboration, or SKU segmentation.

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